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(54) **ELECTRIC MOTOR INCLUDING WINDINGS HAVING CROSSOVERS**

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H02K 3/28 (2006.01)
H02K 15/12 (2006.01)

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See application file for complete search history.

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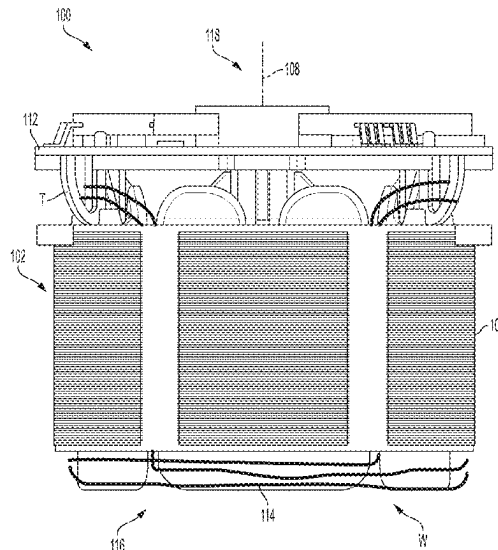
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(57) **ABSTRACT**

A method of arranging windings in a parallel delta configuration in a stator assembly includes connecting a first lead of a first winding to a first terminal. The first terminal is electrically connected to a printed circuit board assembly (PCBA) located at a first axial end of the stator assembly. The method also includes wrapping the first winding about a first tooth of a lamination stack to form a first coil, extending the first winding about at least a portion of a circumference of the lamination stack at a second axial end of the stator assembly toward a second tooth located opposite the first tooth, connecting the first winding to a second terminal electrically connected to the PCBA, wrapping the first winding about the second tooth of the lamination stack to form a second coil, and connecting a second lead of the first winding to the first terminal.

13 Claims, 6 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/698,293, filed on Nov. 27, 2019, now Pat. No. 11,515,743.

- (60) Provisional application No. 62/772,934, filed on Nov. 29, 2018.

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FIG. 1

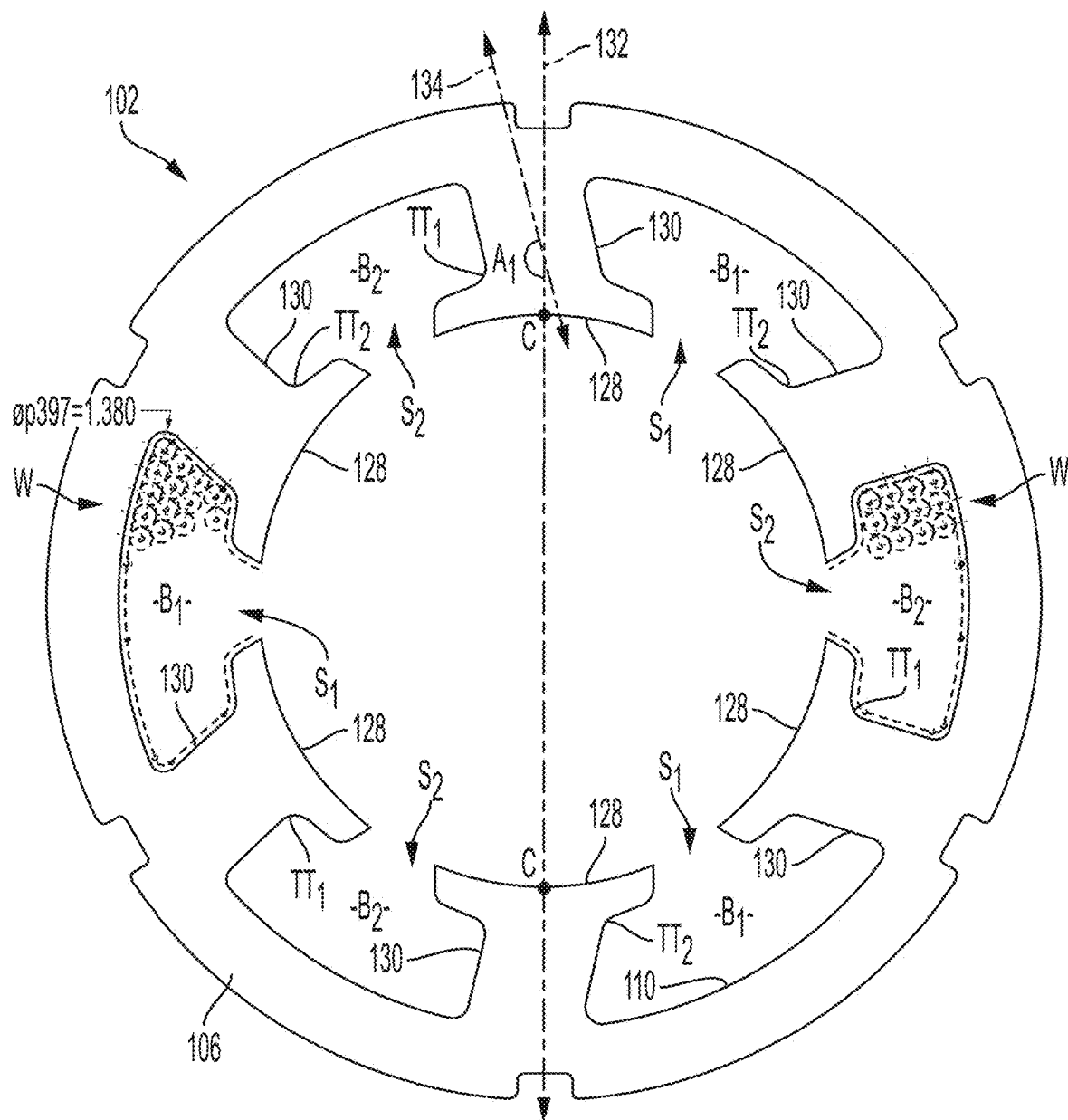


FIG. 2

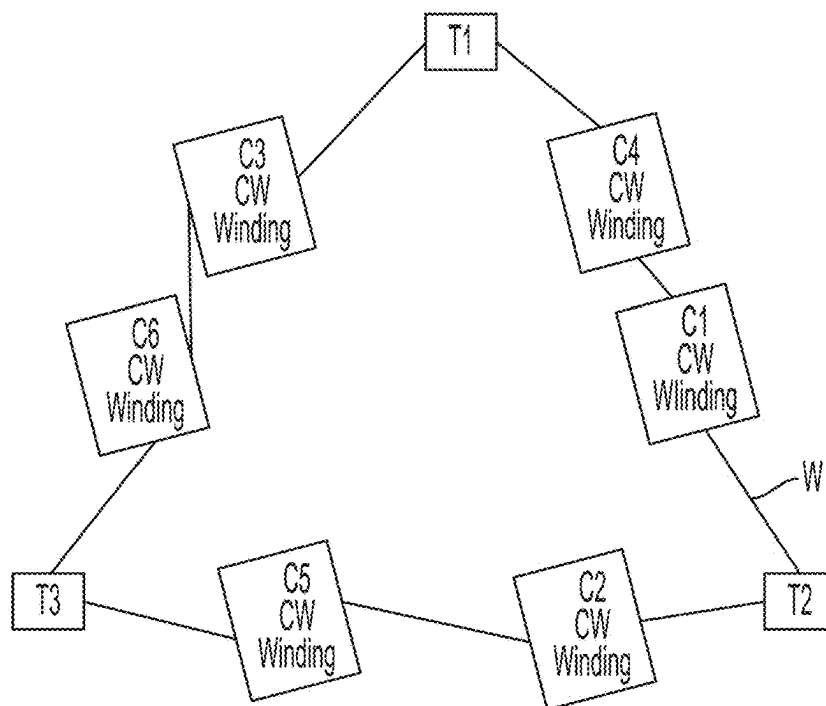


FIG. 3
PRIOR ART

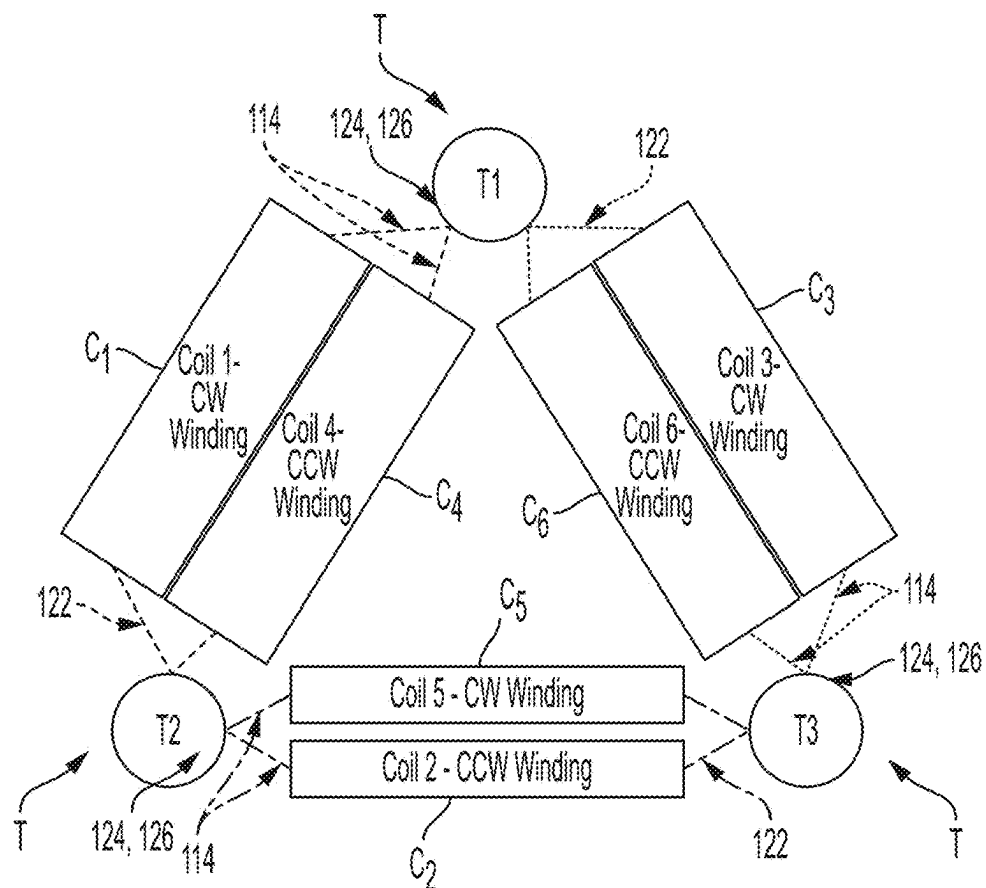


FIG. 4

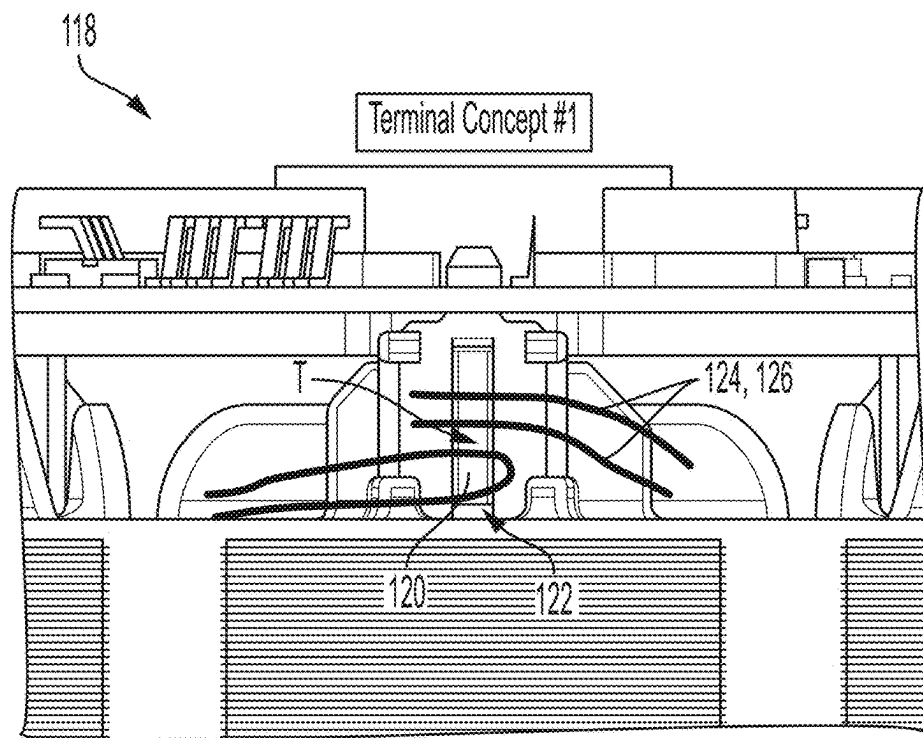


FIG. 5

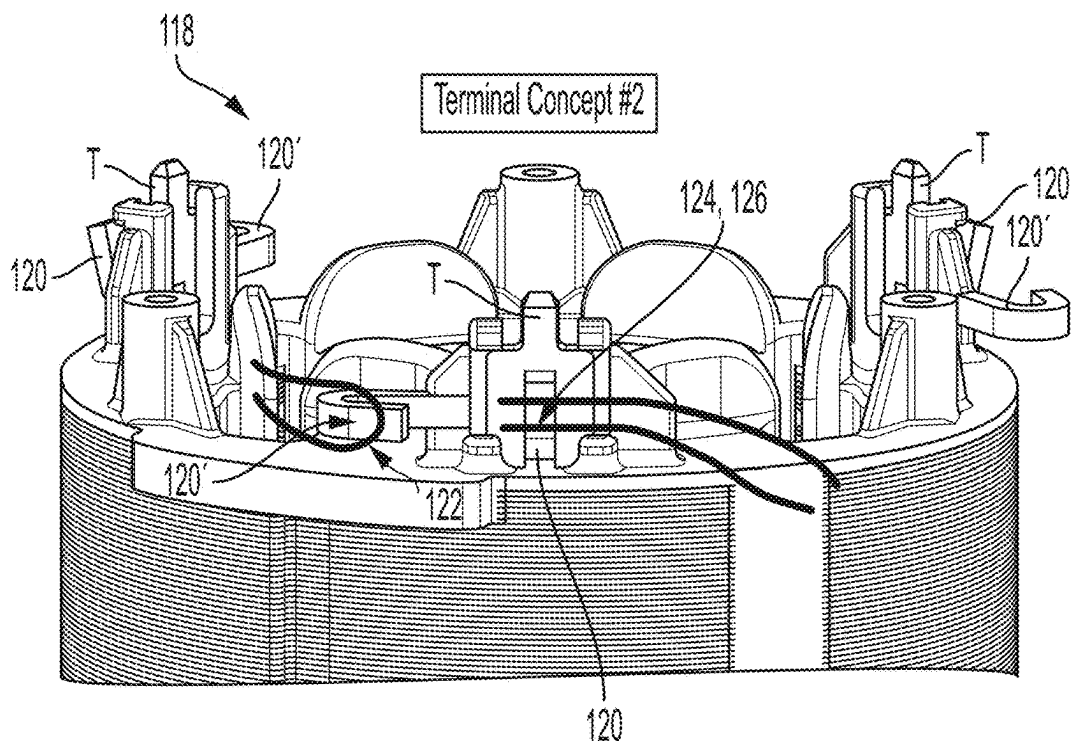
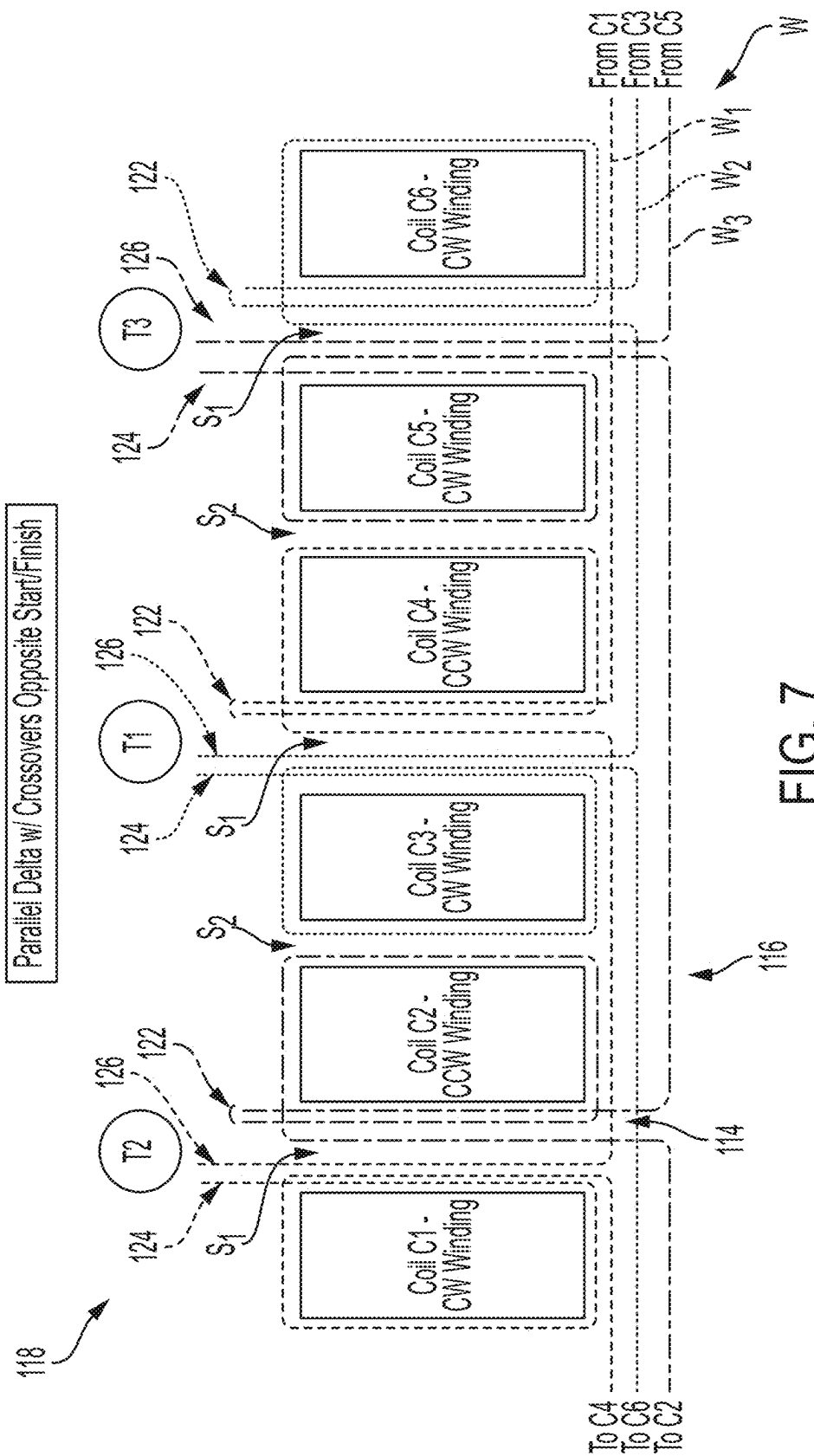


FIG. 6



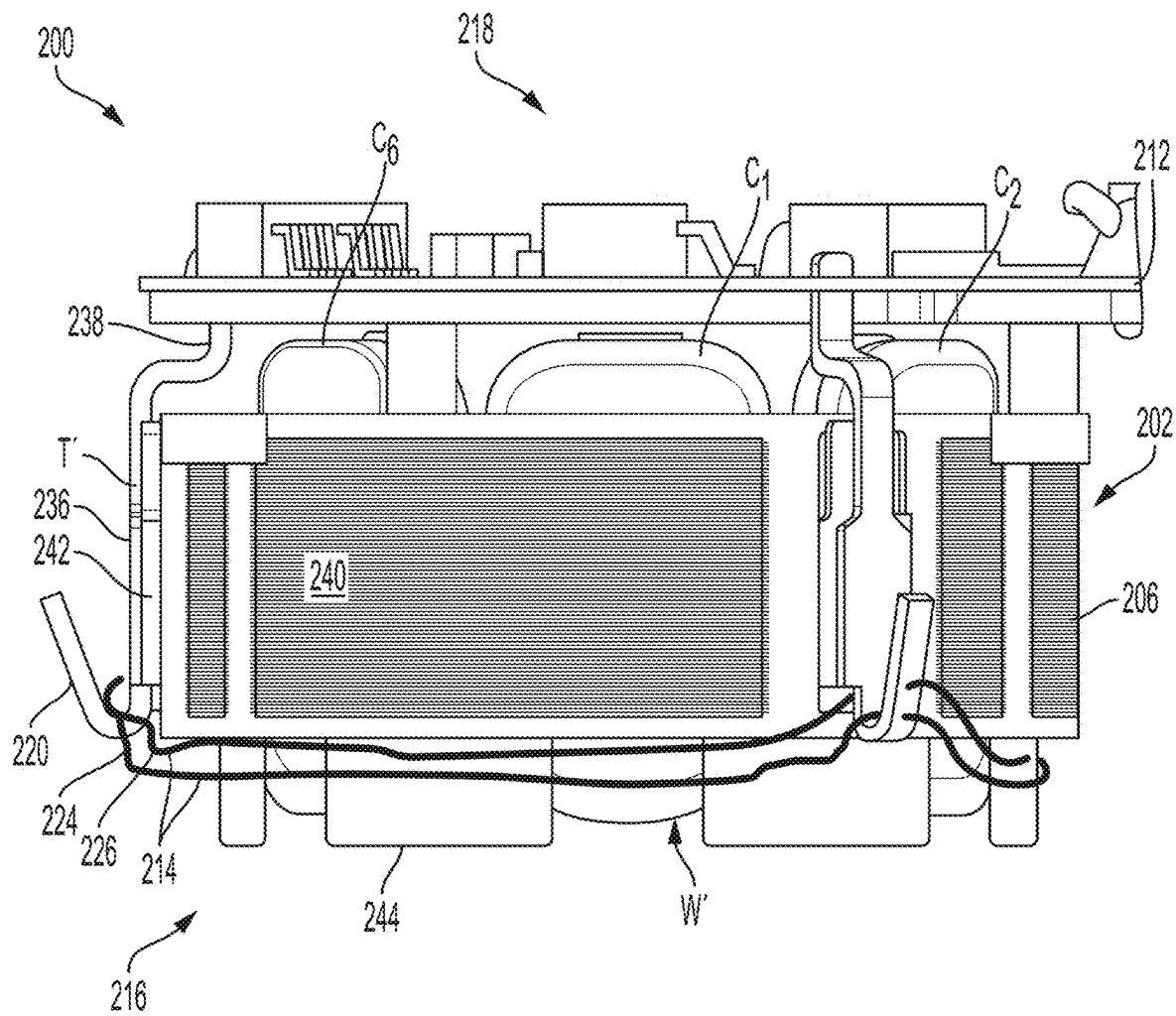


FIG. 8

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ELECTRIC MOTOR INCLUDING WINDINGS HAVING CROSSOVERS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional of U.S. patent application Ser. No. 18/057,844 filed on Nov. 22, 2022, now U.S. Pat. No. 12,003,137, which is a continuation of U.S. patent application Ser. No. 16/698,293 filed on Nov. 27, 2019, now U.S. Pat. No. 11,515,743, which claims priority to U.S. Provisional Patent Application No. 62/772,934 filed on Nov. 29, 2018, the entire contents of each of which is incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates to tools, such as power tools, and more particularly to power tools including an electric motor having windings wound on a stator.

BACKGROUND OF THE INVENTION

Tools, such as power tools, can include an electric motor having a stator to generate a magnetic field. The stator may include windings wound in slots to form coils, and terminals electrically connecting the coils to a power source.

SUMMARY OF THE INVENTION

The present invention includes, in one aspect, a method of arranging a plurality of windings in a parallel delta configuration in a stator assembly. The method includes connecting a first lead of a first winding of the plurality of windings to a first terminal. The first terminal is electrically connected to a printed circuit board assembly (PCBA) located at a first axial end of the stator assembly. The method also includes wrapping the first winding about a first tooth of a lamination stack to form a first coil. The method further includes extending the first winding about at least a portion of a circumference of the lamination stack at a second axial end of the stator assembly toward a second tooth located opposite the first tooth. The method also includes connecting the first winding to a second terminal electrically connected to the PCBA. The method further includes wrapping the first winding about the second tooth of the lamination stack to form a second coil. The method also includes connecting a second lead of the first winding to the first terminal.

The present invention includes, in another aspect, an electric motor including a stator assembly having a lamination stack with an annular yoke and a plurality of teeth extending inwardly from the yoke. The teeth define a plurality of slots therebetween. The plurality of slots include a plurality of first slots and a plurality of second slots. The first slots and the second slots are formed in an alternating sequence in a circumferential direction of the stator assembly. The stator assembly also includes a printed circuit board assembly (PCBA) coupled to the lamination stack. The stator assembly further includes a plurality of terminals electrically connected to the PCBA. The stator assembly also includes a plurality of windings wrapped about the lamination stack to form coils. Each winding includes crossover portions extending about a portion of a circumference of the stator assembly to connect pairs of opposite coils. The windings include leads that contact the terminals to electrically connect the coils to the PCBA. The leads extend

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longitudinally through the first slots only to connect the terminals to the crossover portions.

Other features and aspects of the invention will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a front view of an electric motor according to an embodiment of the invention.

FIG. 2 is a schematic cross-sectional view of a stator assembly of the electric motor of FIG. 1.

FIG. 3 is a schematic diagram of a prior art winding configuration.

FIG. 4 is a schematic diagram of a winding configuration of the electric motor of FIG. 1.

FIG. 5 is a detail view of a portion of the electric motor of FIG. 1.

FIG. 6 is another detail view of a portion of the electric motor of FIG. 1.

FIG. 7 is a schematic diagram of the winding configuration of FIG. 4.

FIG. 8 is a front view of an electric motor according to another embodiment of the invention.

Before any embodiments of the application are explained in detail, it is to be understood that the application is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The application is capable of other embodiments and of being practiced or of being carried out in various ways.

DETAILED DESCRIPTION

FIGS. 1, 2, and 4-7 illustrate an electric motor **100** for use in various different tools, such as power tools (e.g., rotary hammers, pipe threaders, cutting tools, etc.), outdoor tools (e.g., trimmers, pole saws, blowers, etc.), and other electrical devices (e.g., motorized devices, etc.).

The electric motor **100** is configured as a brushless DC motor. In some embodiments, the motor **100** may receive power from an on-board power source (e.g., a battery, not shown). The battery may include any of a number of different nominal voltages (e.g., 12V, 18V, etc.), and may be configured having any of a number of different chemistries (e.g., lithium-ion, nickel-cadmium, etc.). Alternatively, the motor **100** may be powered by a remote power source (e.g., a household electrical outlet) through a power cord. The motor **100** includes a substantially cylindrical stator assembly **102** operable to produce a magnetic field, and a rotor assembly (not shown) supported for rotation with respect to the stator assembly **102**.

The stator assembly **102** includes a lamination stack **106** formed from a plurality of laminations that are stacked along a rotational axis **108**. In the illustrated embodiment, six stator teeth **TT1**, **TT2** (FIG. 2) extend inwardly in a generally radial direction from an annular yoke **110** of the lamination stack **106**. Pairs of adjacent teeth **TT1**, **TT2** form slots **S1**, **S2** therebetween. Although the stator assembly **102** of the illustrated embodiment includes an annular yoke **110**, in other embodiments (not shown) the yoke may be formed in other, non-round shapes (e.g., square, hexagonal, etc.).

Windings **W** (FIG. 1) are routed through the slots **S1**, **S2** (FIG. 2) and wound about the respective teeth **TT1**, **TT2** to form coils **C1**, **C2**, **C3**, **C4**, **C5**, and **C6** (FIG. 7). The windings **W** are electrically connected to terminals **T** (FIG. 1) (e.g., by direct contact), which in turn are electrically

connected to a PCBA 112 (e.g., a hall effect sensor board) that selectively supplies current to the coils C1-C6 via the terminals T.

In some prior art designs as illustrated by FIG. 3, the windings are routed in a series configuration in which a single winding W extends from a first terminal to a first tooth and forms a first coil, then extends from the first coil to a second, opposite tooth and forms a second coil, and then extends from the second coil to a second terminal and attaches thereto. In the proposed electric motor 100 embodied in FIGS. 1, 2, and 4-7, the windings W are configured in a parallel delta configuration (FIG. 4). Specifically, each individual winding W extends between pairs of electrically opposite coils C1-C4, C2-C5, and C3-C6, respectively (FIG. 4). Each pair of opposite coils C1-C4, C2-C5, and C3-C6 is connected in parallel between respective terminals T.

Since the coils of each pair C1-C4, C2-C5, and C3-C6 are located opposite one another, portions of the windings W extend around the circumference of the stator assembly 102 to connect the coils of each pair C1-C4, C2-C5, and C3-C6. These portions of the windings W are referred to herein as crossovers 114 (FIG. 1). In the illustrated embodiment, the crossovers 114 are routed along a fan side 116 of the electric motor 100, and the PCBA 112 is coupled to the motor 100 at a board side 118 of the electric motor 100 opposite the fan side 116. In the illustrated embodiment, the terminals T are coupled to the stator assembly 102 at the board side 118, opposite the crossovers 114. In some embodiments, the electric motor 100 further includes a fan (not shown) rotated by the rotor assembly (not shown), and the fan is located adjacent the fan side 116.

Each terminal T includes a hook portion or tang 120 (FIG. 5), and the windings W are wrapped around the tangs 120 to form the electrical connections between the windings W and the terminals T. This allows the winding configuration to be implemented automatically by a needle winder. The winding configuration also permits the needle winder to wind three of the coils C1-C6 at one time, which can improve the throughput of the winding process. Referring to FIG. 5, the tang 120 extends outward from the stator assembly 102 in a radial direction, and a looped lead 122 of the windings W may be wound around the tang 120 to form the electrical connection between the windings W and the terminal T. In some embodiments, in addition to the tang 120, a second tang 120' may extend away from the terminal T in a circumferential direction as illustrated in FIG. 6.

The windings W may include a first winding W1, a second winding W2, and a third winding W3, each wound about the stator assembly 102 in the configuration illustrated by FIG. 7. Likewise, the terminals T may include a first terminal T1, a second terminal T2, and a third terminal T3. Each of the windings W1, W2, and W3 includes a respective start lead 124 and a respective finish lead 126 that each connect to a respective terminal T1, T2, or T3.

For example, as shown in FIG. 7, the first winding W1 begins at the start lead 124 at the second terminal T2, passes longitudinally through the slot S1, and wraps about the stator assembly 102 to form the first coil C1. The first winding W1 exits the first coil C1 at the fan side 116 and extends about the circumference of the stator assembly 102 (via the crossover 114) toward the location of fourth coil C4. The first winding W1 then extends longitudinally through the slot S1 adjacent the fourth coil C4 to form the looped lead 122 about the tang 120 (FIG. 5) of the first terminal T1, thereby electrically connecting the first winding W1 to the first terminal T1. The first winding W1 then reverses course, passing back through the slot S1 in the longitudinal direc-

tion, and wrapping around the stator assembly 102 to form the fourth coil C4. The first winding W1 exits the fourth coil C4 on the fan side 116, and travels back around the circumference of the stator assembly 102 (via the crossover 114) toward the first coil C1. The first winding W1 then passes longitudinally through the slot S1 adjacent the first coil C1, and terminates at the finish lead 126, which connects to the terminal T2. In this way, the first and fourth coils C1, C4 are connected in parallel between the first and second terminals T1, T2. The second and third windings W2, W3 are wound in a manner similar to the configuration described with respect to W1, as illustrated in FIG. 7.

With continued reference to FIG. 7, the start and finish leads 124, 126 and the looped leads 122 for each winding W1, W2, and W3 all pass longitudinally through slots S1 between the fan side 116 and the board side 118. As a result, each of the slots S1 contains more conductors than each of the slots S2. To account for these extra conductors, the lamination stack 106 is provided with the slots S1 that are larger in area than the slots S2, as will be discussed in further detail below. The larger area of the slots S1 helps to avoid uneven slot fill in the stator assembly 102.

With reference to FIG. 2, each tooth TT1, TT2 of the lamination stack 106 includes a crown portion 128 and a body portion 130 that connects the crown portion 128 to the yoke 110. The crown portions 128 are equally spaced from one another in a circumferential direction. However, the body portions 130 of the teeth TT1, TT2 are alternately offset or skewed in a radial direction, so that each successive tooth TT1, TT2 alternately skews toward the clockwise direction or toward the counter clockwise direction. For example, the teeth TT1 skew slightly clockwise in the circumferential direction, while the teeth TT2 skew slightly counter clockwise in the circumferential direction.

For purposes of illustration, a dividing line 132 divides the lamination stack 106 into equal halves and extends through a center point C of the crown portion 128 of two opposing teeth TT1 and TT2. A longitudinal axis 134 extends through the body portion 130 of the tooth TT1 and intersects the dividing line 132. The dividing line 132 forms an angle A1 with the longitudinal axis 134 of tooth TT1. In the illustrated construction, the angle A1 is approximately 166 degrees. Accordingly, tooth TT1 is skewed or offset at an angle A1 of approximately 166 degrees toward the clockwise direction. Likewise, each of the remaining teeth TT1 are also offset at angles A1 (not shown) of approximately 166 degrees toward the clockwise direction, while the alternating teeth TT2 are offset at corresponding angles A2 (not shown) of approximately 166 degrees toward the counter clockwise direction. In other embodiments, the angles A1, A2 may be greater than 90 degrees and less than 180 degrees.

Because the teeth TT1, TT2 are alternately skewed or offset, the slots S1, S2 defined between adjacent pairs of teeth TT1, TT2 define two different slot areas B1 and B2. The slot area B1 is larger than the slot area B2. In the illustrated embodiment, slot area B1 is approximately 70 millimeters squared (mm²), while slot area B2 is approximately 60 mm². In other embodiments (not shown), the two slot areas B1, B2 may be larger or smaller than 70 mm² and 60 mm², respectively, and may vary as a function of the angles A1, A2. The two different slot areas B1 and B2 help to avoid uneven slot fill when different numbers of windings W are alternately applied to the teeth TT1, TT2.

FIG. 8 illustrates another embodiment of an electric motor 200 similar to the electric motor 100 described above, with like features shown with reference numerals plus "100," or,

prime “()” in the case of alphabetical reference numerals. The electric motor **200** includes stator assembly **202** operable to produce a magnetic field, and a rotor assembly (not shown) supported for rotation with respect to the stator assembly **202**. The stator assembly **202** likewise includes a lamination stack **206** and windings **W'** routed through the slots **S1**, **S2** to form coils **C1'**, **C2'**, **C3'**, **C4'**, **C5'**, and **C6'**. The windings **W'** are electrically connected to terminals **T'**, which in turn are electrically connected to a PCBA **212** (e.g., a hall effect sensor board) that selectively supplies current to the coils **C1'-C6'** via the terminals **T'**.

In the illustrated embodiment, the windings **W'** are configured in a parallel delta configuration wherein each individual winding **W'** extends between pairs of electrically opposite coils **C1'-C4'**, **C2'-C5'**, and **C3'-C6'**, respectively. Each pair of opposite coils **C1'-C4'**, **C2'-C5'**, and **C3'-C6'** is connected in parallel between respective terminals **T'**. Cross-overs **214** are routed along a fan side **216** of the electric motor **200**, and a PCBA **212** is coupled to the electric motor **200** at a board side **218** of the motor **200** opposite the fan side **216**.

The terminals **T'** of the electric motor **200** are elongated such that the terminals **T'** extend longitudinally along an outer surface **240** of the lamination stack **206** between the board side **218** and the fan side **216**. Each elongated terminal **T'** includes a longitudinal portion **236** having a connecting portion **238** electrically coupled to the PCBA **212** at the board side **218** of the electric motor **200**. The longitudinal portion **236** extends along the outer surface **240** of the lamination stack **206** from the connecting portion **238** to a hook portion or tang **220** located adjacent the fan side **216**. Each tang **220** extends along the outer surface **240** of the lamination stack **206** toward the fan side **216**, and then bends backward toward the board side **218** at a location adjacent an axial end of the lamination stack **206**. Start and finish leads **224**, **226** of the windings **W'** connect to the terminals **T'** at the tangs **220** adjacent the fan side **216**. Similarly, looped leads **222** of the windings **W'** also connect to the tangs **220** adjacent the fan side **216**. The elongated terminals **T'** electrically connect the coils **C1'-C6'** to the PCBA **212**.

This configuration allows for significant stack-up reduction to minimize the length of the motor **200** and reduce material and process costs, and also allows the winding configuration to be implemented automatically by a needle winder. The winding configuration also permits the needle winder to wind three of the coils **C1'-C6'** at one time, which can improve the throughput of the winding process. Moreover, the elongated terminals **T'** bridge the length of the stator assembly **202** between the PCBA **212** at the board side **218** and the crossovers **214** at the fan side **216**. Thus, the elongated terminals eliminate the need for the start and finish leads **224**, **226** and the looped leads **222** to pass through the slots **S1'** as described above with respect to the electric motor **100**. This reduces the overall length of the windings **W'** and thereby further reduces material costs.

In the illustrated embodiment, the electric motor **200** includes three elongated terminals **T'** that are spaced apart from one another in the circumferential direction at generally equal intervals. Each terminal **T'** is attached to the outer surface **240** of the lamination stack **206** via an attachment member **242**. The stator assembly **202** also includes an insulating member **244** formed of an insulative material (e.g., plastic) that insulates the windings **W'** from the lamination stack **206**. In the illustrated embodiment, the attachment members **242** are separate elements formed separately from the insulating member **244**. In other embodiments (not shown), the attachment members **242** may be formed as

portions of the insulating member **244**, so that the elongated terminals **T'** are secured to the electric motor **200** via the insulating member **244**.

Although the application has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the application as described.

Various features of the invention are set forth in the following claims.

What is claimed is:

1. A method of arranging a plurality of windings in a parallel delta configuration in a stator assembly, the method comprising:

connecting a first lead of a first winding of the plurality of windings to a first terminal, the first terminal being electrically connected to a printed circuit board assembly (PCBA) located at a first axial end of the stator assembly;

wrapping the first winding about a first tooth of a lamination stack to form a first coil;

extending the first winding about at least a portion of a circumference of the lamination stack at a second axial end of the stator assembly toward a second tooth located opposite the first tooth;

connecting the first winding to a second terminal electrically connected to the PCBA;

wrapping the first winding about the second tooth of the lamination stack to form a second coil; and

connecting a second lead of the first winding to the first terminal.

2. The method of claim 1, wherein after connecting the first lead to the first terminal, the method comprises extending the first winding longitudinally through a first slot located adjacent the first tooth.

3. The method of claim 1, wherein prior to connecting the first winding to the second terminal, the method comprises extending the first winding longitudinally through a second slot located adjacent the second tooth.

4. The method of claim 3, wherein after connecting the first winding to the second terminal, the method comprises extending the first winding longitudinally back through the first slot.

5. The method of claim 1, wherein the first lead connects to the first terminal at the first axial end, and the second lead connects to the first terminal at the first axial end.

6. The method of claim 1, further comprising:

connecting a third lead of a second winding of the plurality of windings to the second terminal;

wrapping the second winding about a third tooth of the lamination stack to form a third coil;

connecting the second winding to a third terminal electrically connected to the PCBA;

wrapping the second winding about a fourth tooth of the lamination stack located opposite the third tooth to form a fourth coil; and

connecting a fourth lead of the second winding to the second terminal.

7. The method of claim 6, further comprising:

connecting a fifth lead of a third winding of the plurality of windings to the third terminal;

wrapping the third winding about a fifth tooth of the lamination stack to form a fifth coil;

connecting the third winding to the first terminal;

wrapping the third winding about a sixth tooth of the lamination stack located opposite the fifth tooth to form a sixth coil; and

connecting a sixth lead of the third winding to the third terminal.

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8. An electric motor comprising:

a stator assembly including

a lamination stack including an annular yoke and a plurality of teeth extending inwardly from the yoke, the teeth defining a plurality of slots therebetween, the plurality of slots comprising a plurality of first slots and a plurality of second slots, the first slots and the second slots being formed in an alternating sequence in a circumferential direction of the stator assembly,

a printed circuit board assembly (PCBA) coupled to the lamination stack,

a plurality of terminals electrically connected to the PCBA; and

a plurality of windings wrapped about the lamination stack to form coils, each winding including crossover portions extending about a portion of a circumference of the stator assembly to connect pairs of opposite coils, the windings including leads that contact the terminals to electrically connect the coils to the PCBA, the leads extending longitudinally through the first slots only to connect the terminals to the crossover portions.

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9. The electric motor of claim **8**, further including a fan located adjacent the crossover portions.

10. The electric motor of claim **8**, wherein each terminal includes a tang, and wherein each winding includes a looped lead that wraps about the tang of at least one terminal to electrically connect the coils to the PCBA.

11. The electric motor of claim **8**, wherein each tooth includes a crown portion and a body portion connecting the crown portion to the yoke, and wherein the body portion of each tooth is skewed away from a radial direction in a circumferential direction, and wherein the body portions of two adjacent teeth of the plurality of teeth alternately skew toward and away from one another in the circumferential direction.

12. The electric motor of claim **8**, wherein each first slot has a first slot area, and each second slot has a second slot area, the first slot area being larger than the second slot area.

13. The electric motor of claim **8**, wherein the stator assembly includes a first axial end and a second axial end, and wherein the PCBA is located at the first axial end and the crossover portions are located at the second axial end.

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